Task 3:

SQL for Data Analysis

```
mysql> CREATE DATABASE ecommerce_analysis;
Query OK, 1 row affected (0.02 sec)

mysql> USE ecommerce_analysis;
Database changed
mysql> CREATE TABLE ecommerce_sales(order_id int, order_date date,customer_id int, product_category varchar(15), region
varchar(15), quantity int, unit_price decimal(10,2), discount decimal(10,2), payment_method varchar(50), delivery_days i
nt, customer_rating decimal(10,2), revenue decimal(10,2));
Query OK, 0 rows affected (0.12 sec)
```

```
mysql> SHOW TABLES;
+------------------------------+
| Tables_in_ecommerce_analysis |
+------------------------------+
| ecommerce_sales              |
+------------------------------+
1 row in set (0.06 sec)

mysql> LOAD DATA LOCAL INFILE 'C:/Users/ASUS/Downloads/ecommerce_sales_analytics_5000.csv' INTO TABLE ecommerce_sales FI
ELDS TERMINATED BY ',' ENCLOSED BY '"' LINES TERMINATED BY '\n' IGNORE 1 ROWS;
Query OK, 5000 rows affected, 5000 warnings (0.13 sec)
Records: 5000  Deleted: 0  Skipped: 0  Warnings: 5000
```

```
mysql> SELECT * FROM ecommerce_sales LIMIT 10;
+----------+------------+-------------+------------------+--------+----------+------------+----------+----------------+--------------
-+-----------------+---------+
| order_id | order_date | customer_id | product_category | region | quantity | unit_price | discount | payment_method | delivery_days
 | customer_rating | revenue |
+----------+------------+-------------+------------------+--------+----------+------------+----------+----------------+--------------
-+-----------------+---------+
|    10001 | 0000-00-00 |        1102 | Beauty           | South  |        7 |     373.65 |     0.28 | Wallet         |            10
 |            4.70 | 1883.20 |
|    10002 | 0000-00-00 |        1435 | Clothing         | South  |        7 |      47.74 |     0.09 | Card           |             6
 |            3.90 |  304.10 |
|    10003 | 0000-00-00 |        1860 | Beauty           | East   |        3 |     311.28 |     0.31 | COD            |             6
 |            2.50 |  644.35 |
|    10004 | 0000-00-00 |        1270 | Electronics      | West   |        5 |     524.47 |     0.02 | Wallet         |             6
 |            1.60 | 2569.90 |
|    10005 | 0000-00-00 |        1106 | Clothing         | West   |        5 |     139.87 |     0.33 | Wallet         |             4
 |            4.90 |  468.56 |
|    10006 | 0000-00-00 |        1071 | Electronics      | West   |        2 |     264.44 |     0.20 | Card           |             3
 |            3.30 |  423.10 |
|    10007 | 0000-00-00 |        1700 | Clothing         | South  |        6 |     564.60 |     0.23 | Card           |             1
 |            1.80 | 2608.45 |
|    10008 | 0000-00-00 |        1020 | Beauty           | North  |        6 |     265.71 |     0.13 | Card           |             2
 |            3.70 | 1387.01 |
|    10009 | 0000-00-00 |        1614 | Clothing         | South  |        6 |     224.65 |     0.27 | Card           |             2
 |            1.40 |  983.97 |
|    10010 | 0000-00-00 |        1121 | Clothing         | West   |        4 |     272.71 |     0.06 | Card           |            10
 |            4.50 | 1025.39 |
+----------+------------+-------------+------------------+--------+----------+------------+----------+----------------+--------------
-+-----------------+---------+
10 rows in set (0.00 sec)
```

```
mysql> SELECT COUNT(*) AS Total_Records
    -> FROM ecommerce_sales;
+---------------+
| Total_Records |
+---------------+
|          5000 |
+---------------+
1 row in set (0.04 sec)

mysql> SELECT order_id, product_category, revenue
    -> FROM ecommerce_sales
    -> LIMIT 10;
+----------+------------------+---------+
| order_id | product_category | revenue |
+----------+------------------+---------+
|    10001 | Beauty           | 1883.20 |
|    10002 | Clothing         |  304.10 |
|    10003 | Beauty           |  644.35 |
|    10004 | Electronics      | 2569.90 |
|    10005 | Clothing         |  468.56 |
|    10006 | Electronics      |  423.10 |
|    10007 | Clothing         | 2608.45 |
|    10008 | Beauty           | 1387.01 |
|    10009 | Clothing         |  983.97 |
|    10010 | Clothing         | 1025.39 |
+----------+------------------+---------+
10 rows in set (0.00 sec)
```

```
mysql> SELECT *
    -> FROM ecommerce_sales
    -> WHERE product_category = 'Electronics';
+----------+------------+-------------+------------------+--------+----------+------------+----------+----------------+--------------
-+-----------------+---------+
| order_id | order_date | customer_id | product_category | region | quantity | unit_price | discount | payment_method | delivery_days
 | customer_rating | revenue |
+----------+------------+-------------+------------------+--------+----------+------------+----------+----------------+--------------
-+-----------------+---------+
|    10004 | 0000-00-00 |        1270 | Electronics      | West   |        5 |     524.47 |     0.02 | Wallet         |             6
 |            1.60 | 2569.90 |
|    10006 | 0000-00-00 |        1071 | Electronics      | West   |        2 |     264.44 |     0.20 | Card           |             3
 |            3.30 |  423.10 |
|    10012 | 0000-00-00 |        1214 | Electronics      | West   |        3 |     384.71 |     0.14 | COD            |            11
 |            1.90 |  992.55 |
|    10018 | 0000-00-00 |        1871 | Electronics      | East   |        7 |     399.90 |     0.29 | COD            |             3
 |            3.80 | 1987.50 |
|    10020 | 0000-00-00 |        1130 | Electronics      | East   |        7 |     341.46 |     0.23 | Card           |            11
 |            3.20 | 1840.47 |
|    10026 | 0000-00-00 |        1413 | Electronics      | South  |        7 |      25.73 |     0.22 | Card           |             8
 |            2.00 |  140.49 |
|    10028 | 0000-00-00 |        1385 | Electronics      | North  |        7 |     170.23 |     0.14 | Card           |             8
 |            4.60 | 1024.78 |
|    10030 | 0000-00-00 |        1955 | Electronics      | East   |        6 |     323.83 |     0.14 | COD            |             2
 |            4.10 | 1670.96 |
|    10031 | 0000-00-00 |        1276 | Electronics      | North  |        6 |     141.32 |     0.20 | Card           |             1
 |            1.70 |  678.34 |
|    10033 | 0000-00-00 |        1459 | Electronics      | North  |        6 |      64.94 |     0.09 | Wallet         |             9
 |            3.50 |  354.57 |
|    10035 | 0000-00-00 |        1021 | Electronics      | West   |        2 |     511.91 |     0.19 | COD            |             9
 |            1.50 |  829.29 |
|    10036 | 0000-00-00 |        1252 | Electronics      | West   |        4 |     375.27 |     0.02 | COD            |            11
 |            3.90 | 1471.06 |
```

```
mysql> SELECT *
    -> FROM ecommerce_sales
    -> ORDER BY revenue DESC
    -> LIMIT 10;
+----------+------------+-------------+------------------+--------+----------+------------+----------+----------------+--------------
-+-----------------+---------+
| order_id | order_date | customer_id | product_category | region | quantity | unit_price | discount | payment_method | delivery_days
 | customer_rating | revenue |
+----------+------------+-------------+------------------+--------+----------+------------+----------+----------------+--------------
-+-----------------+---------+
|    10810 | 0000-00-00 |        1890 | Electronics      | South  |        7 |     594.42 |     0.01 | Card           |             4
 |            4.90 | 4119.33 |
|    14884 | 0000-00-00 |        1146 | Electronics      | South  |        7 |     583.31 |     0.00 | Card           |             5
 |            4.40 | 4083.17 |
|    12215 | 0000-00-00 |        1113 | Electronics      | South  |        7 |     591.15 |     0.02 | Wallet         |             1
 |            3.20 | 4055.29 |
|    13102 | 0000-00-00 |        1699 | Beauty           | East   |        7 |     596.12 |     0.04 | COD            |             6
 |            3.00 | 4005.93 |
|    13687 | 0000-00-00 |        1831 | Clothing         | North  |        7 |     578.59 |     0.03 | COD            |             8
 |            1.60 | 3928.63 |
|    10158 | 0000-00-00 |        1779 | Electronics      | North  |        7 |     571.07 |     0.02 | COD            |            10
 |            4.50 | 3917.54 |
|    13350 | 0000-00-00 |        1082 | Electronics      | West   |        7 |     574.15 |     0.03 | Card           |            10
 |            2.70 | 3898.48 |
|    10322 | 0000-00-00 |        1608 | Electronics      | West   |        7 |     585.43 |     0.05 | Card           |            10
 |            3.20 | 3893.11 |
|    10662 | 0000-00-00 |        1550 | Clothing         | East   |        7 |     581.67 |     0.05 | COD            |             3
 |            1.20 | 3868.11 |
|    14306 | 0000-00-00 |        1283 | Beauty           | South  |        7 |     555.09 |     0.02 | COD            |            11
 |            4.90 | 3807.92 |
+----------+------------+-------------+------------------+--------+----------+------------+----------+----------------+--------------
```

```
mysql> SELECT SUM(revenue) AS Total_Revenue
    -> FROM ecommerce_sales;
+---------------+
| Total_Revenue |
+---------------+
|    5109775.74 |
+---------------+
1 row in set (0.03 sec)

mysql> SELECT AVG(customer_rating) AS Average_Rating
    -> FROM ecommerce_sales;
+----------------+
| Average_Rating |
+----------------+
|       2.973980 |
+----------------+
1 row in set (0.01 sec)

mysql> SELECT SUM(quantity) AS Total_Quantity
    -> FROM ecommerce_sales;
+----------------+
| Total_Quantity |
+----------------+
|          20224 |
+----------------+
1 row in set (0.01 sec)
```

```
mysql> SELECT
    -> product_category,
    -> SUM(revenue) AS Category_Revenue
    -> FROM ecommerce_sales
    -> GROUP BY product_category;
+------------------+------------------+
| product_category | Category_Revenue |
+------------------+------------------+
| Beauty           |        765860.88 |
| Clothing         |       1531931.72 |
| Electronics      |       1829899.22 |
| Home             |        982083.92 |
+------------------+------------------+
4 rows in set (0.07 sec)

mysql> SELECT
    -> region,
    -> SUM(revenue) AS Region_Revenue
    -> FROM ecommerce_sales
    -> GROUP BY region;
+--------+----------------+
| region | Region_Revenue |
+--------+----------------+
| South  |     1246640.90 |
| East   |     1236044.23 |
| West   |     1345582.16 |
| North  |     1281508.45 |
+--------+----------------+
4 rows in set (0.01 sec)
```

JOINS

```
mysql> drop table customer_details;
Query OK, 0 rows affected (0.06 sec)

mysql> create table customer_details(customer_id int, customer_name varchar(15), customer_age int, gender varchar(15));
Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO customer_details VALUES(1082, 'Arun', 25, 'Male'),(1108, 'Rahul', 30, 'Male'),(1357, 'Anu', 28, 'Female'),(1541, '
Meera', 35, 'Female');
Query OK, 4 rows affected (0.03 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM customer_details;
+-------------+---------------+--------------+--------+
| customer_id | customer_name | customer_age | gender |
+-------------+---------------+--------------+--------+
|        1082 | Arun          |           25 | Male   |
|        1108 | Rahul         |           30 | Male   |
|        1357 | Anu           |           28 | Female |
|        1541 | Meera         |           35 | Female |
+-------------+---------------+--------------+--------+
4 rows in set (0.00 sec)
```

```
mysql> select e.order_id,e.customer_id,e.product_category,e.revenue,c.customer_name,c.gender from ecommerce_sales e inner join custom
er_details c on e.customer_id = c.customer_id;
+----------+-------------+------------------+---------+---------------+--------+
| order_id | customer_id | product_category | revenue | customer_name | gender |
+----------+-------------+------------------+---------+---------------+--------+
|    10818 |        1108 | Electronics      |  359.61 | Rahul         | Male   |
|    10953 |        1357 | Clothing         | 1690.63 | Anu           | Female |
|    11044 |        1082 | Clothing         |  682.23 | Arun          | Male   |
|    11115 |        1541 | Electronics      |  796.49 | Meera         | Female |
|    11267 |        1108 | Electronics      |  850.55 | Rahul         | Male   |
|    11821 |        1108 | Electronics      |  812.49 | Rahul         | Male   |
|    12812 |        1357 | Electronics      |  323.47 | Anu           | Female |
|    13184 |        1541 | Electronics      | 1193.50 | Meera         | Female |
|    13350 |        1082 | Electronics      | 3898.48 | Arun          | Male   |
|    13784 |        1108 | Electronics      | 2011.59 | Rahul         | Male   |
|    13786 |        1541 | Beauty           |   69.37 | Meera         | Female |
|    14136 |        1108 | Clothing         |  338.23 | Rahul         | Male   |
|    14285 |        1541 | Home             | 2126.82 | Meera         | Female |
|    14299 |        1082 | Electronics      |  798.57 | Arun          | Male   |
|    14520 |        1108 | Electronics      |  757.83 | Rahul         | Male   |
|    14719 |        1541 | Electronics      | 1216.04 | Meera         | Female |
|    14742 |        1108 | Beauty           |  119.87 | Rahul         | Male   |
|    14762 |        1108 | Electronics      |  289.94 | Rahul         | Male   |
|    14928 |        1541 | Electronics      | 2050.61 | Meera         | Female |
|    14961 |        1357 | Electronics      |  511.84 | Anu           | Female |
|    14974 |        1082 | Clothing         | 1135.54 | Arun          | Male   |
|    14976 |        1357 | Electronics      | 2613.30 | Anu           | Female |
|    14986 |        1357 | Clothing         |  739.00 | Anu           | Female |
+----------+-------------+------------------+---------+---------------+--------+
23 rows in set (0.03 sec)
```

```
mysql> SELECT
    -> e.order_id,
    -> e.customer_id,
    -> c.customer_name,
    -> e.revenue
    -> FROM ecommerce_sales e
    -> LEFT JOIN customer_details c
    -> ON e.customer_id = c.customer_id;\
+----------+-------------+---------------+---------+
| order_id | customer_id | customer_name | revenue |
+----------+-------------+---------------+---------+
|    10001 |        1102 | NULL          | 1883.20 |
|    10002 |        1435 | NULL          |  304.10 |
|    10003 |        1860 | NULL          |  644.35 |
|    10004 |        1270 | NULL          | 2569.90 |
|    10005 |        1106 | NULL          |  468.56 |
|    10006 |        1071 | NULL          |  423.10 |
|    10007 |        1700 | NULL          | 2608.45 |
|    10008 |        1020 | NULL          | 1387.01 |
|    10009 |        1614 | NULL          |  983.97 |
|    10010 |        1121 | NULL          | 1025.39 |
|    10011 |        1466 | NULL          | 1875.01 |
|    10012 |        1214 | NULL          |  992.55 |
|    10013 |        1330 | NULL          | 2368.71 |
|    10014 |        1458 | NULL          | 1447.71 |
|    10015 |        1087 | NULL          | 1617.79 |
```

```
mysql> SELECT
    -> c.customer_id,
    -> c.customer_name,
    -> e.order_id,
    -> e.revenue
    -> FROM ecommerce_sales e
    -> RIGHT JOIN customer_details c
    -> ON e.customer_id = c.customer_id;
+-------------+---------------+----------+---------+
| customer_id | customer_name | order_id | revenue |
+-------------+---------------+----------+---------+
|        1082 | Arun          |    14974 | 1135.54 |
|        1082 | Arun          |    14299 |  798.57 |
|        1082 | Arun          |    13350 | 3898.48 |
|        1082 | Arun          |    11044 |  682.23 |
|        1108 | Rahul         |    14762 |  289.94 |
|        1108 | Rahul         |    14742 |  119.87 |
|        1108 | Rahul         |    14520 |  757.83 |
|        1108 | Rahul         |    14136 |  338.23 |
|        1108 | Rahul         |    13784 | 2011.59 |
|        1108 | Rahul         |    11821 |  812.49 |
|        1108 | Rahul         |    11267 |  850.55 |
|        1108 | Rahul         |    10818 |  359.61 |
|        1357 | Anu           |    14986 |  739.00 |
|        1357 | Anu           |    14976 | 2613.30 |
|        1357 | Anu           |    14961 |  511.84 |
|        1357 | Anu           |    12812 |  323.47 |
|        1357 | Anu           |    10953 | 1690.63 |
|        1541 | Meera         |    14928 | 2050.61 |
|        1541 | Meera         |    14719 | 1216.04 |
|        1541 | Meera         |    14285 | 2126.82 |
|        1541 | Meera         |    13786 |   69.37 |
|        1541 | Meera         |    13184 | 1193.50 |
```

SUBQUERIES

```
mysql> select order_id,customer_id,revenue from ecommerce_sales where revenue > (select avg(revenue) from ecommerce_sales);
+----------+-------------+---------+
| order_id | customer_id | revenue |
+----------+-------------+---------+
|    10001 |        1102 | 1883.20 |
|    10004 |        1270 | 2569.90 |
|    10007 |        1700 | 2608.45 |
|    10008 |        1020 | 1387.01 |
|    10010 |        1121 | 1025.39 |
|    10011 |        1466 | 1875.01 |
|    10013 |        1330 | 2368.71 |
|    10014 |        1458 | 1447.71 |
|    10015 |        1087 | 1617.79 |
|    10018 |        1871 | 1987.50 |
|    10020 |        1130 | 1840.47 |
|    10025 |        1491 | 1423.15 |
|    10027 |        1805 | 2559.59 |
|    10028 |        1385 | 1024.78 |
|    10029 |        1191 | 1048.56 |
|    10030 |        1955 | 1670.96 |
|    10032 |        1160 | 2715.34 |
|    10036 |        1252 | 1471.06 |
|    10038 |        1856 | 1081.12 |
|    10040 |        1474 | 3640.56 |
|    10041 |        1058 | 1440.47 |
|    10043 |        1681 | 1581.23 |
|    10046 |        1975 | 2534.44 |
|    10051 |        1957 | 2663.13 |
|    10055 |        1243 | 1737.34 |
|    10058 |        1130 | 1910.37 |
|    10059 |        1484 | 2796.31 |
|    10065 |        1273 | 1791.05 |
```

```
mysql> select * from ecommerce_sales where revenue = (select max(revenue) from ecommerce_sales);
+----------+------------+-------------+------------------+--------+----------+------------+----------+----------------+--------------
-+-----------------+---------+
| order_id | order_date | customer_id | product_category | region | quantity | unit_price | discount | payment_method | delivery_days
 | customer_rating | revenue |
+----------+------------+-------------+------------------+--------+----------+------------+----------+----------------+--------------
-+-----------------+---------+
|    10810 | 0000-00-00 |        1890 | Electronics      | South  |        7 |     594.42 |     0.01 | Card           |             4
 |            4.90 | 4119.33 |
+----------+------------+-------------+------------------+--------+----------+------------+----------+----------------+--------------
-+-----------------+---------+
1 row in set (0.02 sec)
```

VIEWS

```
mysql> CREATE VIEW revenue_analysis AS
    -> SELECT
    -> product_category,
    -> region,
    -> SUM(revenue) AS Total_Revenue,
    -> AVG(customer_rating) AS Average_Rating
    -> FROM ecommerce_sales
    -> GROUP BY product_category, region;
Query OK, 0 rows affected (0.02 sec)

mysql> select * from revenue_analysis;
+------------------+--------+---------------+----------------+
| product_category | region | Total_Revenue | Average_Rating |
+------------------+--------+---------------+----------------+
| Beauty           | South  |     191073.98 |       2.965698 |
| Clothing         | South  |     348363.14 |       2.966575 |
| Beauty           | East   |     174909.11 |       2.952727 |
| Electronics      | West   |     458589.26 |       2.968778 |
| Clothing         | West   |     416326.55 |       3.034606 |
| Beauty           | North  |     186213.32 |       3.172680 |
| Clothing         | East   |     402582.60 |       3.066582 |
| Clothing         | North  |     364659.43 |       2.947354 |
| Electronics      | East   |     411803.35 |       2.906938 |
| Home             | West   |     257001.88 |       2.862214 |
| Home             | North  |     258184.77 |       3.025207 |
| Electronics      | South  |     487055.68 |       2.987253 |
| Electronics      | North  |     472450.93 |       2.938528 |
| Home             | East   |     246749.17 |       2.949799 |
| Home             | South  |     220148.10 |       2.930556 |
| Beauty           | West   |     213664.47 |       2.930729 |
+------------------+--------+---------------+----------------+
16 rows in set (0.04 sec)
```

Optimize queries with indexes

```
mysql> CREATE INDEX idx_customer_id
    -> ON ecommerce_sales(customer_id);
Query OK, 0 rows affected (0.14 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> SHOW INDEX FROM ecommerce_sales;
+-----------------+------------+-----------------+--------------+-------------+-----------+-------------+----------+--------+------+-
-----------+---------+---------------+---------+------------+
| Table           | Non_unique | Key_name        | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null |
Index_type | Comment | Index_comment | Visible | Expression |
+-----------------+------------+-----------------+--------------+-------------+-----------+-------------+----------+--------+------+-
-----------+---------+---------------+---------+------------+
| ecommerce_sales |          1 | idx_customer_id |            1 | customer_id | A         |         989 |     NULL |   NULL | YES  |
BTREE      |         |               | YES     | NULL       |
+-----------------+------------+-----------------+--------------+-------------+-----------+-------------+----------+--------+------+-
-----------+---------+---------------+---------+------------+
1 row in set (0.04 sec)
```